# Yunxiao LIU

yunxiao.liu@proton.me | linkedin.com/in/yunxiao-liu | github.com/AlbertRyu | yunxiao.de | +49 015236380991

## Education

**Technische Universität München (TUM)** Munich, Germany
*M.Sc. Applied and Engineering Physics* *2023 – Oct. 2026*

- Average grade: 1.5 | Theoretical Solid State Physics (1.0), Computational Methods in Many-Body Physics (1.7)
- Thesis (in progress; expected graduation: Sep. 2026): Structural and magnetic properties in halogen-functionalized hybrid organic-inorganic compounds $(4\,X\text{−BA})_2\text{MnCl}_4$ ($X$ = H, Cl, I, Br)

**University of Macau** Macau, China
*B.Sc. Applied Physics and Chemistry* *2019 – 2023*

- Cumulative grade: 3.84/4 (German grade equivalent: 1.1) | Dean's Honour List (2019–2023)
- ICBC Macau Scholarship (2021/2022); Frank Wong Scholarship (2019/2020)
- Thesis: Two-dimensional metal oxides: a first-principles study

## Research Experience

**Working Student** Apr. 2024 – Sep. 2025
*Forschungs-Neutronenquelle Heinz Maier-Leibnitz (FRM II)* *Munich, Germany*

- Supervisor: Dr. Jitae Park; Dr. Lukas Beddrich
- Investigated magnetic phase transitions of layered hybrid perovskite materials (4X-Mn-BA series, X = Cl, Br, I, H, F)
- Characterised magnetic and structural properties using polarized microscope, Quantum Design PPMS, and both single-crystal and powder X-ray diffraction
- Developed Python library to automate processing and visualization of PPMS magnetometry data

**Undergraduate Research Assistant** Oct. 2021 – Jun. 2023
*Institute of Applied Physics and Materials Engineering* *Macau, China*

- Supervisor: Dr. Hui Pan
- Synthesized novel water-splitting photocatalysts via hydrothermal methods, screening candidate materials for optimal catalytic performance
- Characterized materials using SEM, TEM, and electrochemical workstation to evaluate morphology, structure, and catalytic activity
- Performed first-principles calculations (VASP/DFT) to model electronic properties and thermodynamic stability of candidate materials

## Projects

**pySpinW** | *Python, Open Source, Scientific Computing* 2026 – present

- Contributor and member of the pySpinW development team for SpinW for Python, an open-source package for magnon calculations and scattering simulations
- Contributed code to the project and collaborated with maintainers on extending the Python ecosystem for spin-wave modelling

**PPMS_ToolKit** | *Python, PySide6, SQLite, Parquet, matplotlib* 2024 – present

- Developed a pip-installable library for managing and visualizing Quantum Design PPMS data (VSM, Heat Capacity)
- Stores imported `.dat` files in SQLite with Parquet-backed caching for fast repeated loading and deduplication
- Ships a Qt6 GUI with customizable plot legends, curve toggling, and automatic ZFC/FC mode detection from filenames

**vsm_visualizer** | *Python, DearPyGui* 2025

- Lightweight desktop viewer for PPMS VSM data: split-panel UI with a file browser and live matplotlib preview, distributed as a standalone `.exe` requiring no installation

**Personal Homelab** | *Raspberry Pi 5, Docker, Cloudflare Tunnel, Linux* 2025 – present

- Self-hosted productivity services (Seafile, Immich, Paperless-ngx) on Raspberry Pi 5
- Exposed services securely via Cloudflare Tunnel without opening inbound firewall ports

## Technical Skills

**Languages**: English (C1), German (B1), Mandarin Chinese (Native)
**Experimental Techniques**:VSM magnetometry, Single Crystal/Powder XRD, polarized light microscopy

## Interests

**Bouldering**: regularly climb around Fontainebleau grade 5–6; enjoy training at Boulderwelt OST and DAV Thalkirchen
**Reading**: especially science fiction; have read most of Isaac Asimov and prefer stories with strong world-building and storytelling
**AI and Cognition**: exploring how AI can fit into work and daily life while preserving and strengthening human critical thinking